

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	07 Jul 2026
Project Name	SHOPEZ E-commerce Application
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Sign-up Form Input validation for Name, Email & Password User account creation in MongoDB
FR-2	User Authentication	Login using Email & Password JWT Token Generation Secure Logout
FR-3	Product Management	View Product List Search Products View Product Details Browse Products by Category
FR-4	Shopping Features	Add Products to Cart Update Cart Quantity Remove Products from Cart Add/Remove Products from Wishlist
FR-5	Order Management	Place Order Track order View Order History Display Order Confirmation
FR-6	User Interface	Responsive Design Navigation Bar Error & Validation Messages

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should have a simple, responsive, and user-friendly interface that is easy to navigate.

NFR-2	Security	User passwords must be securely stored, and JWT authentication should protect user sessions and API endpoints.
NFR-3	Reliability	The application should operate consistently with minimal errors and recover gracefully from failures.
NFR-4	Performance	Product pages and API responses should load within 2–3 seconds under normal usage.
NFR-5	Availability	The application should be accessible 24×7 with minimal downtime during deployment or maintenance.
NFR-6	Scalability	The system should support increasing numbers of users, products, and orders without significant performance degradation.